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Convolutional Neural Networks

Importar Librerías

In [285]:
```
import numpy as np
import os
import re
import matplotlib.pyplot as plt
%matplotlib inline
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
```

In [286]:
```
import tensorflow as tf
import keras
from keras.utils import to_categorical
# Importación de Input para definir la entrada del modelo: keras es una API pública y estable
from keras import Input
# from tensorflow.python.keras.layers import Input (usa una ruta interna de TensorFlow)
from keras.models import Sequential, Model
from keras.layers import Dense, Dropout, Flatten
from keras.layers import Conv2D, MaxPooling2D
from keras.layers import BatchNormalization
from keras.layers import LeakyReLU
```

Cargar set de Imágenes

In [287]:

```
# # # # # #
# Conexión con Google Drive acceso a imágenes de entrenamiento .../sportimages/golf
# Conexión con Google Drive acceso a imágenes de entrenamiento .../sportimages/basket
# Conexión con Google Drive acceso a imágenes de pruebas .../test_images
from google.colab import drive
import os

drive.mount("/content/drive", force_remount=True)
```

```
Mounted at /content/drive
```

In [288]:

```
# Ruta original del script
dirname = os.path.join(os.getcwd(), 'drive/MyDrive/Colab_Notebooks/sportimages')
imgpath = dirname + os.sep

images = []
directories = []
dircount = []
prevRoot=''
cant=0

print("leyendo imagenes de ",imgpath)

for root, dirnames, filenames in os.walk(imgpath):
    for filename in filenames:
        if re.search(r"\.(jpg|jpeg|png|bmp|tiff)$", filename):
            cant=cant+1
            filepath = os.path.join(root, filename)
            image = plt.imread(filepath)
            images.append(image)
            b = "Leyendo..." + str(cant)
            print (b, end="\r")
            if prevRoot !=root:
                print(root, cant)
                prevRoot=root
                directories.append(root)
                dircount.append(cant)
                cant=0
dircount.append(cant)

dircount = dircount[1:]
# Evita que se rompa el código si no encuentra el dataset
if len(dircount) > 0:
    dircount[0]=dircount[0]+1
# dircount[0]=dircount[0]+1
print('Directorios leidos:',len(directories))
print("Imagenes en cada directorio", dircount)
print('suma Total de imagenes en subdirs:',sum(dircount))
```

```
leyendo imagenes de  /content/drive/MyDrive/Colab_Notebooks/sportimages/
/content/drive/MyDrive/Colab_Notebooks/sportimages/golf 1
/content/drive/MyDrive/Colab_Notebooks/sportimages/basket 9768
Directorios leidos: 2
Imagenes en cada directorio [9769, 1854]
suma Total de imagenes en subdirs: 11623
```

Creamos las etiquetas

In [289]:

```
labels=[]
indice=0
for cantidad in dircount:
    for i in range(cantidad):
        labels.append(indice)
    indice=indice+1
print("Cantidad etiquetas creadas: ",len(labels))
```

```
Cantidad etiquetas creadas:  11623
```

In [290]:

```
deportes=[]
indice=0
for directorio in directories:
    name = directorio.split(os.sep)
    print(indice , name[len(name)-1])
    deportes.append(name[len(name)-1])
    indice=indice+1
```

```
0 golf
1 basket
```

In [291]:

```
y = np.array(labels)
X = np.array(images, dtype=np.uint8) #convierto de lista a numpy

# Find the unique numbers from the train labels
classes = np.unique(y)
nClasses = len(classes)
print('Total number of outputs : ', nClasses)
print('Output classes : ', classes)
```

```
Total number of outputs :  2
Output classes :  [0 1]
```

Creamos Sets de Entrenamiento y Test

In [292]:

```
train_X,test_X,train_Y,test_Y = train_test_split(X,y,test_size=0.2)
print('Training data shape : ', train_X.shape, train_Y.shape)
print('Testing data shape : ', test_X.shape, test_Y.shape)
```

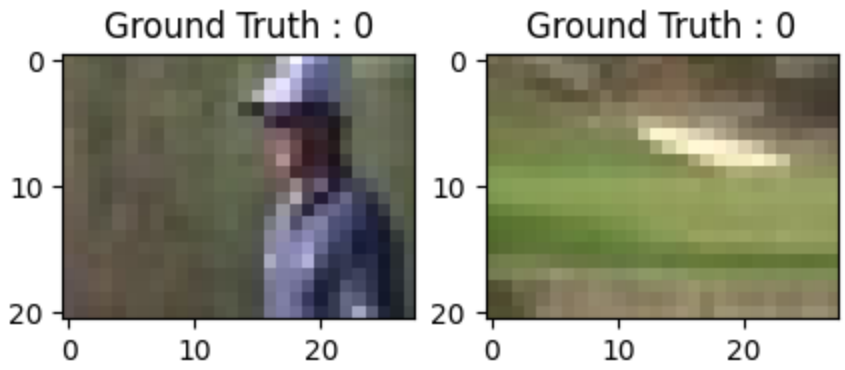
```
Training data shape :  (9298, 21, 28, 3) (9298,)
Testing data shape :  (2325, 21, 28, 3) (2325,)
```

In [293]:
```
plt.figure(figsize=[5,5])

# Display the first image in training data
plt.subplot(121)
plt.imshow(train_X[0,:,:], cmap='gray')
plt.title("Ground Truth : {}".format(train_Y[0]))

# Display the first image in testing data
plt.subplot(122)
plt.imshow(test_X[0,:,:], cmap='gray')
plt.title("Ground Truth : {}".format(test_Y[0]))
```

Out[293]: Text(0.5, 1.0, 'Ground Truth : 0')



Preprocesamos las imagenes

In [294]:
```
train_X = train_X.astype('float32')
test_X = test_X.astype('float32')
train_X = train_X / 255.
test_X = test_X / 255.
```

Hacemos el One-hot Encoding para la red

In [295]:
```
# Change the labels from categorical to one-hot encoding
train_Y_one_hot = to_categorical(train_Y)
test_Y_one_hot = to_categorical(test_Y)

# Display the change for category label using one-hot encoding
print('Original label:', train_Y[0])
print('After conversion to one-hot:', train_Y_one_hot[0])
```

```
Original label: 0
After conversion to one-hot: [1. 0.]
```

Creamos el Set de Entrenamiento y Validación

In [296]:
```
#Mezclar todo y crear los grupos de entrenamiento y testing
train_X,valid_X,train_label,valid_label = train_test_split(train_X, train_Y_one_hot, test_size=0.2, random_state=13)
```

In [297]:
```
print(train_X.shape,valid_X.shape,train_label.shape,valid_label.shape)
```

```
(7438, 21, 28, 3) (1860, 21, 28, 3) (7438, 2) (1860, 2)
```

Creamos el modelo de CNN

In [298]:
```
#declaramos variables con los parámetros de configuración de la red
INIT_LR = 1e-3 # Valor inicial de learning rate. El valor 1e-3 corresponde con 0.001
epochs = 6 # Cantidad de iteraciones completas al conjunto de imagenes de entrenamiento
batch_size = 64 # cantidad de imágenes que se toman a la vez en memoria
```

In [299]:
```
sport_model = Sequential()
sport_model.add(Input(shape=(21,28,3)))
sport_model.add(Conv2D(32, kernel_size=(3, 3), activation='linear', padding='same'))
# sport_model.add(Conv2D(32, kernel_size=(3, 3),activation='linear',padding='same',input_shape=(21,28,3)))
sport_model.add(LeakyReLU(negative_slope=0.1))
sport_model.add(MaxPooling2D((2, 2),padding='same'))
sport_model.add(Dropout(0.5))

sport_model.add(Flatten())
sport_model.add(Dense(32, activation='linear'))
sport_model.add(LeakyReLU(negative_slope=0.1))
sport_model.add(Dropout(0.5))
sport_model.add(Dense(nClasses, activation='softmax'))
```

In [300]:
```
sport_model.summary()
```

Model: "sequential_10"

Layer (type)	Output Shape	Param #
conv2d_10 (Conv2D)	(None, 21, 28, 32)	896
leaky_re_lu_20 (LeakyReLU)	(None, 21, 28, 32)	0
max_pooling2d_10 (MaxPooling2D)	(None, 11, 14, 32)	0
dropout_20 (Dropout)	(None, 11, 14, 32)	0
flatten_10 (Flatten)	(None, 4928)	0
dense_20 (Dense)	(None, 32)	157,728
leaky_re_lu_21 (LeakyReLU)	(None, 32)	0
dropout_21 (Dropout)	(None, 32)	0
dense_21 (Dense)	(None, 2)	66

Total params: 158,690 (619.88 KB)

Trainable params: 158,690 (619.88 KB)

Non-trainable params: 0 (0.00 B)

In [301]:
```
sport_model.compile(loss=keras.losses.categorical_crossentropy,
optimizer=keras.optimizers.Adagrad(learning_rate=INIT_LR, weight_decay=INIT_LR / 100),metrics=['accuracy'])
```

Entrenamos el modelo: Aprende a clasificar imágenes

In [302]:

```
# este paso puede tomar varios minutos, dependiendo de tu ordenador, cpu y memoria ram libre
sport_train = sport_model.fit(
    train_X,
    train_label,
    batch_size=batch_size,
    epochs=epochs,
    verbose=1,
    validation_data=(valid_X, valid_label)
)
# sport_train = sport_model.fit(train_X, train_label, batch_size=batch_size, epochs=epochs, verbose=1, validation_data=
(valid_X, valid_label))
```

```
Epoch 1/6
[1m117/117[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m6s[0m 24ms/step - accuracy: 0.8232 - loss: 0.4588 -
val_accuracy: 0.8457 - val_loss: 0.3925
Epoch 2/6
[1m117/117[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 4ms/step - accuracy: 0.8399 - loss: 0.4097 -
val_accuracy: 0.8457 - val_loss: 0.3675
Epoch 3/6
[1m117/117[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 4ms/step - accuracy: 0.8392 - loss: 0.3901 -
val_accuracy: 0.8457 - val_loss: 0.3419
Epoch 4/6
[1m117/117[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 4ms/step - accuracy: 0.8408 - loss: 0.3629 -
val_accuracy: 0.8457 - val_loss: 0.3171
Epoch 5/6
[1m117/117[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 4ms/step - accuracy: 0.8469 - loss: 0.3432 -
val_accuracy: 0.8543 - val_loss: 0.2902
Epoch 6/6
[1m117/117[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m1s[0m 4ms/step - accuracy: 0.8513 - loss: 0.3219 -
val_accuracy: 0.8688 - val_loss: 0.2683
```

In [303]:

```
# guardamos la red, para reutilizarla en el futuro, sin tener que volver a entrenar
#sport_model.save("sports_mnist.h5py")
# sport_model.save("sports_mnist.keras")
sport_model.save("/content/drive/MyDrive/Colab_Notebooks/keras/sports_mnist.keras")
```

Evaluamos la red

In [304]:

```
test_eval = sport_model.evaluate(test_X, test_Y_one_hot, verbose=1)
```

```
[1m73/73[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m1s[0m 7ms/step - accuracy: 0.8615 - loss: 0.2735
```

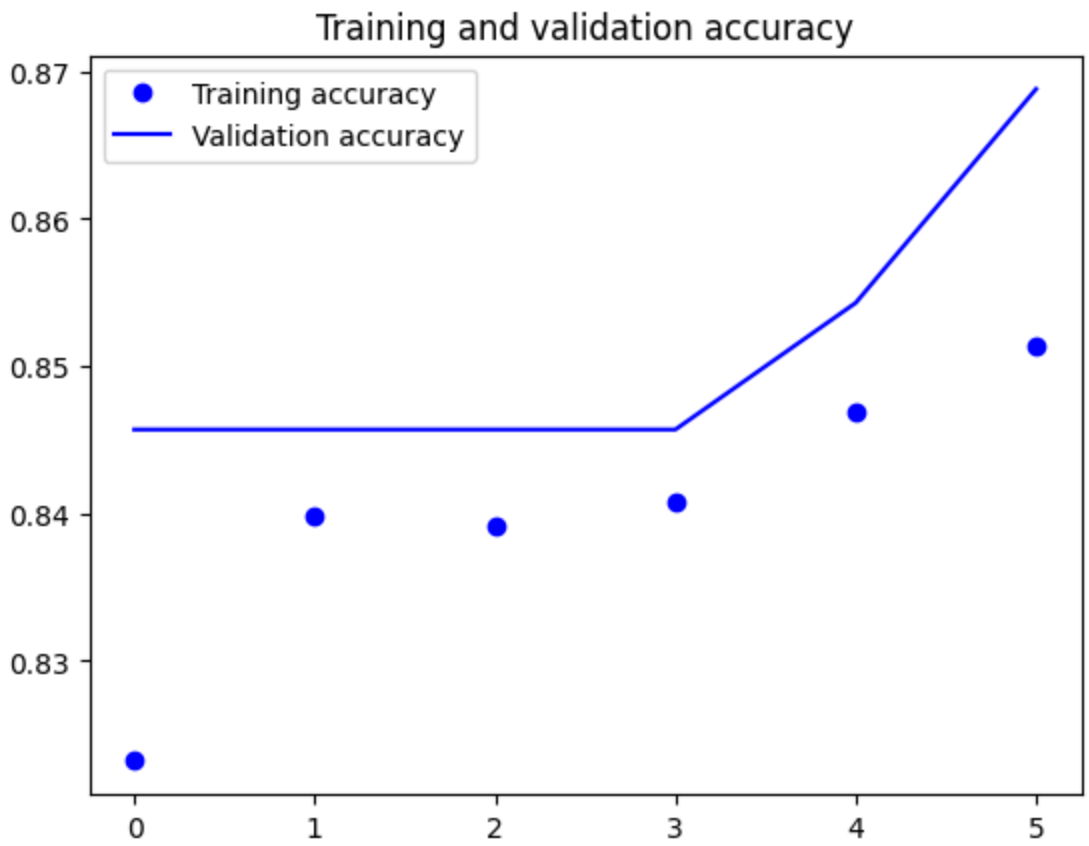
In [305]:

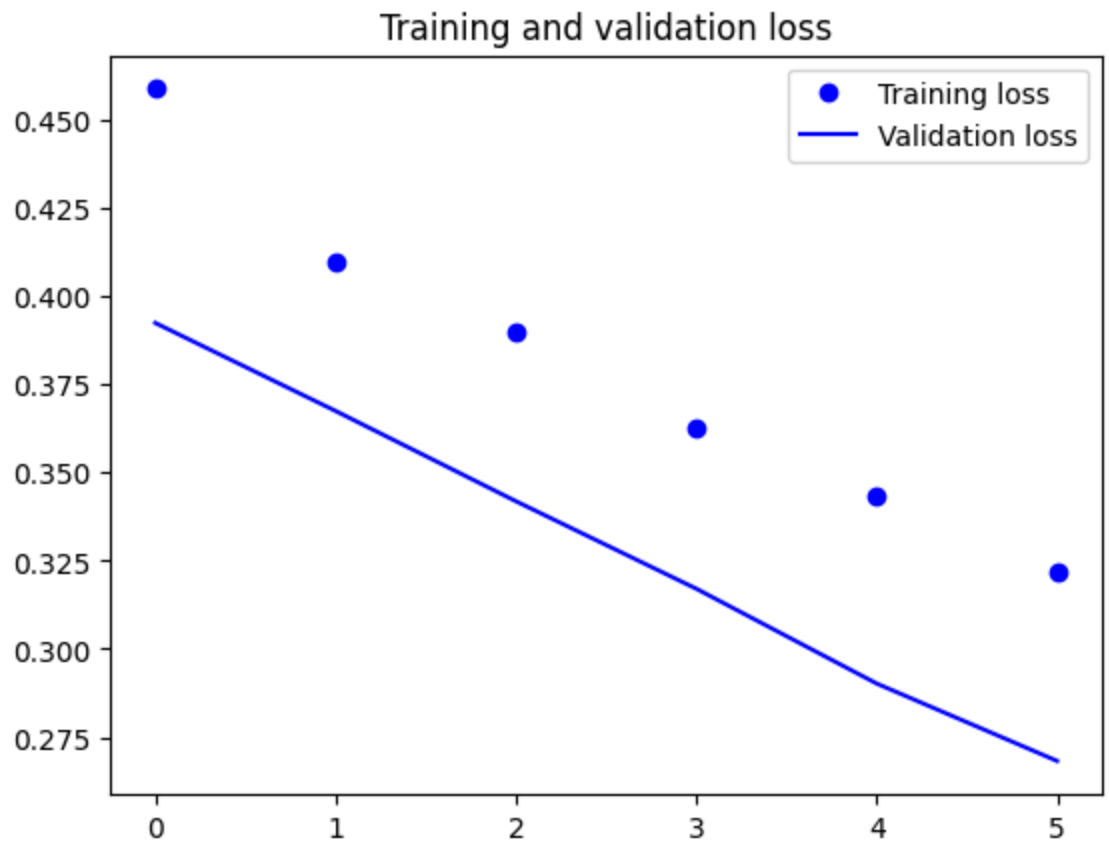
```
print('Test loss:', test_eval[0])
print('Test accuracy:', test_eval[1])
```

```
Test loss: 0.27350831031799316
Test accuracy: 0.861505389213562
```

In [306]:

```
accuracy = sport_train.history['accuracy']
val_accuracy = sport_train.history['val_accuracy']
loss = sport_train.history['loss']
val_loss = sport_train.history['val_loss']
epochs = range(len(accuracy))
plt.plot(epochs, accuracy, 'bo', label='Training accuracy')
plt.plot(epochs, val_accuracy, 'b', label='Validation accuracy')
plt.title('Training and validation accuracy')
plt.legend()
plt.figure()
plt.plot(epochs, loss, 'bo', label='Training loss')
plt.plot(epochs, val_loss, 'b', label='Validation loss')
plt.title('Training and validation loss')
plt.legend()
plt.show()
```


Training and validation accuracy
Training accuracy
Validation accuracy
0.87
0.86
0.85
0.84
0.83
0
1
2
3
4
5



In [307]:
```
predicted_classes2 = sport_model.predict(test_X)
```

```
[1m73/73[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m1s[0m 5ms/step
```

In [308]:
```
predicted_classes=[]
for predicted_sport in predicted_classes2:
    predicted_classes.append(predicted_sport.tolist().index(max(predicted_sport)))
predicted_classes=np.array(predicted_classes)
```

In [309]:

```
predicted_classes.shape, test_Y.shape
```

Out[309]: ((2325,), (2325,))

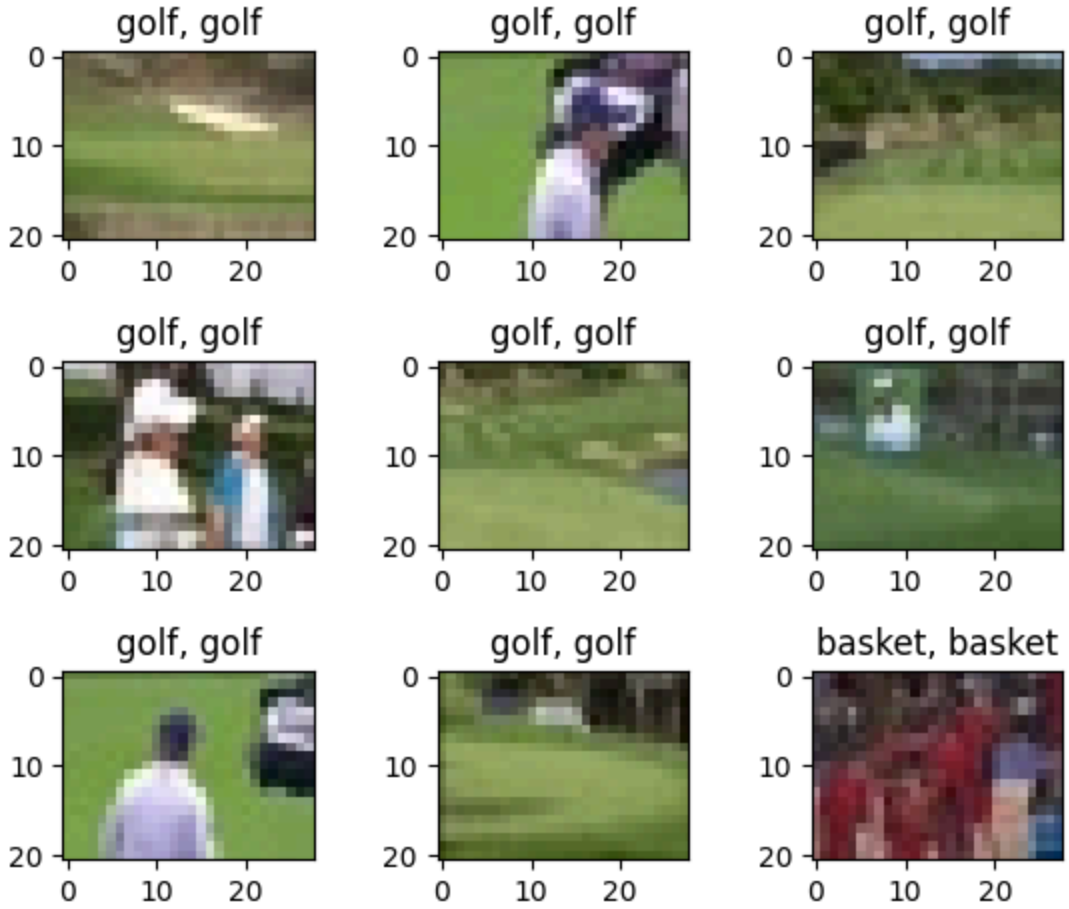
Aprendamos de los errores: Qué mejorar

In [310]:

```
correct = np.where(predicted_classes==test_Y)[0]
print("Found %d correct labels" % len(correct))
for i, correct in enumerate(correct[0:9]):
    plt.subplot(3,3,i+1)
    plt.imshow(test_X[correct].reshape(21,28,3), cmap='gray', interpolation='none')
    plt.title("{}, {}".format(deportes[predicted_classes[correct]],
                                                    deportes[test_Y[correct]]))

    plt.tight_layout()
```

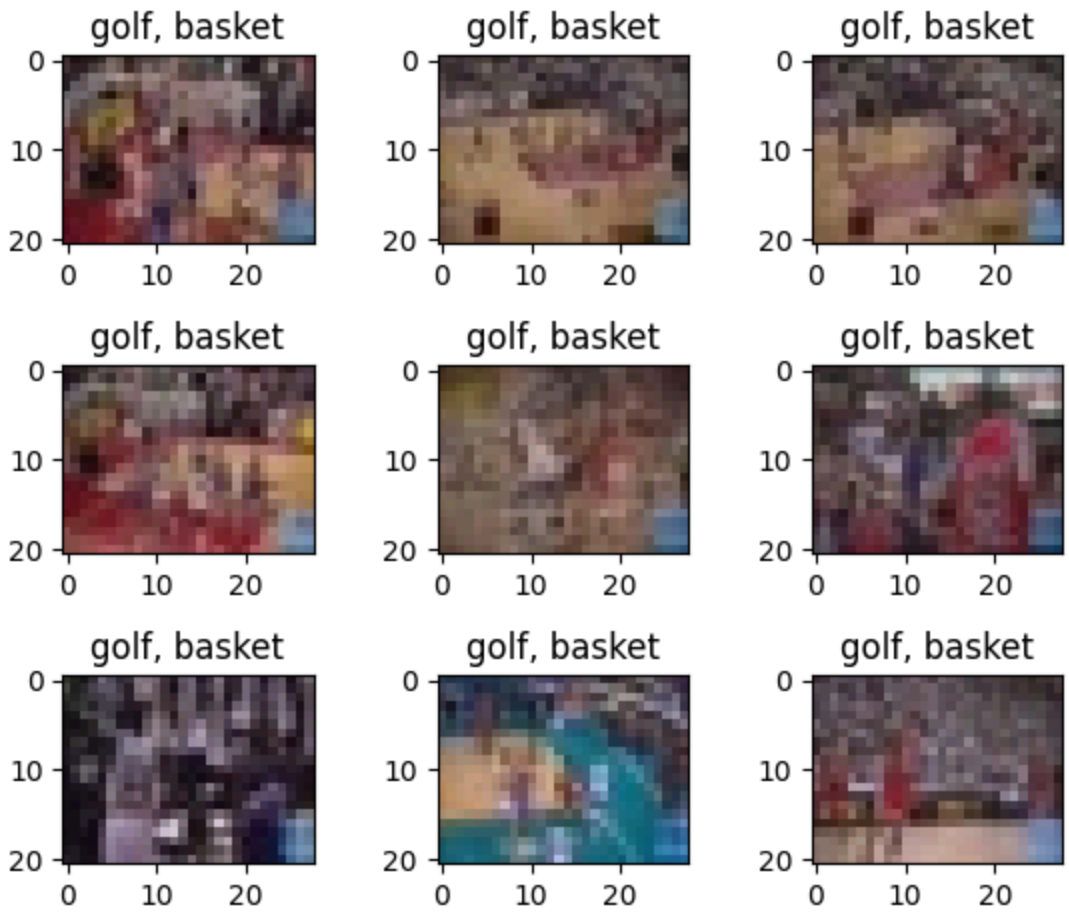
```
Found 2003 correct labels
```


golf, golf
golf, golf
golf, golf
golf, golf
golf, golf
golf, golf
golf, golf
golf, golf
basket, basket
0
10
20
0
10
20

In [311]:

```
incorrect = np.where(predicted_classes!=test_Y)[0]
print("Found %d incorrect labels" % len(incorrect))
for i, incorrect in enumerate(incorrect[0:9]):
    plt.subplot(3,3,i+1)
    plt.imshow(test_X[incorrect].reshape(21,28,3), cmap='gray', interpolation='none')
    plt.title("{}, {}".format(deportes[predicted_classes[incorrect]],
                                                        deportes[test_Y[incorrect]]))
    plt.tight_layout()
```

```
Found 322 incorrect labels
```



In [312]:

```
target_names = ["Class {}".format(i) for i in range(nClasses)]
# print(classification_report(test_Y, predicted_classes, target_names=target_names))
print(classification_report(test_Y, predicted_classes, target_names=target_names, zero_division=0))
```

```
              precision    recall  f1-score   support

     Class 0       0.86      1.00      0.92      1951
     Class 1       0.91      0.16      0.26       374

    accuracy                           0.86      2325
   macro avg       0.88      0.58      0.59      2325
weighted avg       0.87      0.86      0.82      2325
```

Prediccion de una nueva imagen

In [313]:

```
from skimage.transform import resize

images = []
# AQUI ESPECIFICAMOS UNAS IMAGENES
filenames = ['/content/drive/MyDrive/Colab_Notebooks/test_images/golf_0466.jpg']


for filepath in filenames:
    image = plt.imread(filepath,0)
    image_resized = resize(image, (21, 28), anti_aliasing=True, clip=False, preserve_range=True)
    images.append(image_resized)

X = np.array(images, dtype=np.uint8) #convierto de lista a numpy
test_X = X.astype('float32')
test_X = test_X / 255.

predicted_classes = sport_model.predict(test_X)

for i, img_tagged in enumerate(predicted_classes):
    print(filenames[i], deportes[img_tagged.tolist().index(max(img_tagged))])
```

```
[1m1/1[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m1s[0m 649ms/step
/content/drive/MyDrive/Colab_Notebooks/test_images/golf_0466.jpg golf
```

In [313]:

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